

# Nitrogen and carbon incorporation in GaN<sub>x</sub>As<sub>1-x</sub> grown in a showerhead MOVPE reactor

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## ABSTRACT

The growth of dilute nitrides with metal organic chemical vapor phase epitaxy (MOVPE) presents two major challenges: a low N-incorporation combined with a high impurity density of carbon and hydrogen. For the device implementation the full understanding of the growth of the metastable material and the influence of the parameters on the impurity levels is of high interest. We investigated the N- and C-incorporation in GaN<sub>x</sub>As<sub>1-x</sub> as a reference system in a close coupled showerhead MOVPE reactor to minimize gas-phase reactions for a wide range of growth parameters and precursors. The N-incorporation was always proportional to the N/As-ratio, while the absolute N-molar-flow, growth rate and interaction time show only a minor influence, indicating little gas-phase pre-reactions. Pyrolysis and pre-reactions between radicals on the surface have been taken into account according to the used precursor combination. We identified the group-III-methyl-groups as the major C-source and found a linear correlation between the incorporated N and C. GaN<sub>0.01</sub>As<sub>0.99</sub> could be grown with C-background below 10<sup>16</sup> atoms/cm<sup>3</sup> for optimized conditions.

## 1. Introduction

Dilute nitrides have quite unique properties among the III-V compound semiconductors concerning their bandgap and lattice constant. Adding small amounts of nitrogen (N) to a host crystal like GaAs significantly reduces the bandgap. This effect has been described in the literature with a band anti-crossing model [1]. The combination with other elements like indium (In) or antimony (Sb) allows bandgap engineering in the range from 0.9 to 1.4 eV without changing the lattice constant, offering an easy implementation on GaAs- or Ge-substrates. For this reason, dilute nitrides are ideally suited for applications such as quantum well lasers [2–4] or multi-junction solar cells [5–7].

However, despite their advantages, there has been no wide use of these materials in industrial devices. This can be correlated to two major reasons: First of all, the epitaxial growth of dilute nitrides requires extreme non-equilibrium conditions. Secondly, the material exhibits poor opto-electronic properties. Notably, the latter prevents many applications and can be correlated to point defects originating from the N or high impurity densities of carbon (C) or hydrogen (H) atoms. Hence, understanding the growth mechanism, particularly the N-incorporation, and its influence on the impurity density is of major interest.

Alloying GaAs with N creates a metastable crystal growth regime, as N has a significant smaller covalent radius compared to the host crystal [8]. Hence, extreme non-equilibrium growth conditions have to be applied to incorporate even small amounts of N, while higher N-concentrations cannot be grown due to a large miscibility gap. Typical growth temperatures are usually in the range of 500 to 600 °C for metal-organic vapor phase epitaxy, with a N-incorporation from the gas to the solid phase sometimes lower than 1% [9]. This low N-incorporation can be attributed to two major causes: first, low decomposition efficiencies of the precursor molecules due to low growth temperatures [10], chemical pre-reactions between the precursors [11] and losses of the active N-species in chemical reactions [12]; second, enhanced desorption of the active N-species, most probably NH<sub>2</sub>, from the surface. The latter effect can be modified by either changing the strain state of the crystal [9] or exchanging the group-III element to aluminum (Al) or indium (In), where the N-incorporation is enhanced or lowered, respectively, due to differences in the local crystal bond strength compared to GaAs [9]. As most of these processes occur in parallel, a detailed picture of the growth remains a challenge, despite the intensive research in the past years.

We chose to face the challenge of investigating the growth of GaNAs

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in a closed coupled showerhead reactor. The specific reactor design with a distance between the inlet of the gases and the wafer surface of less than 20 mm minimizes pre-reactions in the gas-phase, allowing a partial separation of the above described effects. Focusing on the ternary material GaNAs simplifies the interpretation of the results, while the results should still be applicable to the technologically more interesting quaternary materials such as GaInNAs which can be grown lattice-matched to GaAs. The simple reference system allows an in-depth investigation of the influence of growth parameters like temperature, gas-phase ratios or growth rate. Additionally, we varied the Gallium- (Ga) and N-precursors to extend the investigated parameter space. Thus, interactions between the precursors and their decomposition products can be monitored. The performed study allows a parallel investigation of the C-incorporation into the material, giving a more complete picture of the growth process and offering paths to lower the impurity level for future applications. Hence, this paper is separated in two parts, starting with the discussion of the N-incorporation into GaAs as a function of different growth parameters, followed by the investigation of the C-incorporation into GaNAs alloy.

## 2. Experimental procedure

All samples were grown on (001)-GaAs-substrates with an offset of  $6^\circ$  towards the [111]-direction in a CRIUS closed coupled showerhead (CSS) metal organic vapor phase epitaxy reactor with a  $7 \times 4''$  wafer configuration in a 100 mbar H-atmosphere. We used the precursors triethylgallium (TEGa) and trimethylgallium (TMGa) on the group III side and tertiarybutylarsine (TBAs), unsymmetric dimethylhydrazine (UDMHy) and tertiarybutylhydrazine (TBHy) on the group V side. An EpiTT from Laytec was used for in-situ control of the surface temperature. The N-content and layer thickness were determined by high resolution X-ray diffraction (HR-XRD) rocking curves of the (004)-reflex in a Xpert MRD Pro from Panalytical. Quantitative C-concentrations were measured with secondary ion mass spectroscopy (SIMS) using ion-implanted reference samples. To assure a homogenous temperature profile the center recess of the  $7 \times 4''$  configuration was not used. The wafers were put on one of the 6 recess points around the center wafer for which the temperature difference across the wafer is below 1.5 K. Compositional variation of N on the wafers was below 0.1% N in  $\text{GaN}_x\text{As}_{1-x}$  with a N-content between 0.5 and 2.5%. This result was confirmed by several test samples using XRD measurements. The SIMS and XRD analysis for this work were always performed within the center area of the wafers.

For N-incorporation-tests we used a  $3x \text{ GaN}_x\text{As}_{1-x} / \text{GaAs}$  multi-quantum well structure with thicknesses of approximately 10 and 70 nm, respectively, to avoid relaxation of the GaNAs-layer. Fig. 1 shows an example measurement with the corresponding simulation, including the used parameters. The SIMS measurements were performed on 250 nm thick layers to achieve stable levels during the measurement. The measurements were performed at RTG Mikroanalysis in Berlin using C and N implantation standards in GaAs for quantitative results. The GaAs-Carbon background was below  $2 \times 10^{15} \text{ cm}^{-3}$  for all measurement.

## 3. Results and discussion

The aim of this paper is to report the results of the investigation of N- and unintentional C-incorporation into  $\text{GaN}_x\text{As}_{1-x}$  in a MOVPE showerhead reactor. Hence, this section is separated into two parts: The first one presents the influence of various growth parameters on the N-incorporation. Especially, the effect of the N/As ratio, growth temperature, As/III ratio, distance between showerhead and susceptor, as well as, growth rate have been investigated for different gallium and nitrogen precursors. Finally, a comparison to other reactors is made. The second part discusses the influence of the previously investigated growth parameters on the C-background in GaNAs.

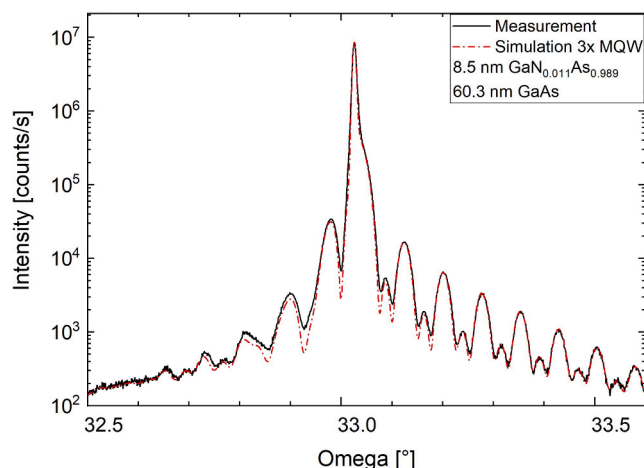


Fig. 1. HR-XRD rocking curve of the (004)-reflex of a  $3x \text{ GaN}_x\text{As}_{1-x} / \text{GaAs}$  multi-quantum-well structure (black) on GaAs (001) substrate. The red dashed-dotted line shows the corresponding simulation with the included parameters. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

### 3.1. N-Composition

In this first section, the results of the influence of the above mentioned growth parameters on the N-incorporation will be presented. The effects of the different growth parameters are discussed step-by-step, to develop a comprehensive picture of the GaNAs growth.

#### 3.1.1. Temperature and N-precursor

First of all, the influence of the gas-phase ratio and temperature on the N-incorporation for different N-precursors is investigated. Fig. 2 shows the N-concentration,  $x$ , in  $\text{GaN}_x\text{As}_{1-x}$  as a function of the N/As-ratio of the precursors in the gas-phase. It is obvious that the N incorporation for all N-precursors and temperature combinations, which have been investigated, shows a linear dependence of the N incorporation with the N/As-ratio, rather than with the actual gas-phase ratio  $\text{N}/(\text{As} + \text{N})$ . This trend is in good agreement with the literature [9,12,13]. It should be emphasized that approximately only 1% N is incorporated into the crystal for a gas phase ratio of 1, indicating a loss of 98% of the available precursor molecules. As mentioned above, this low incorporation efficiency is caused by the volatility of the N decomposition

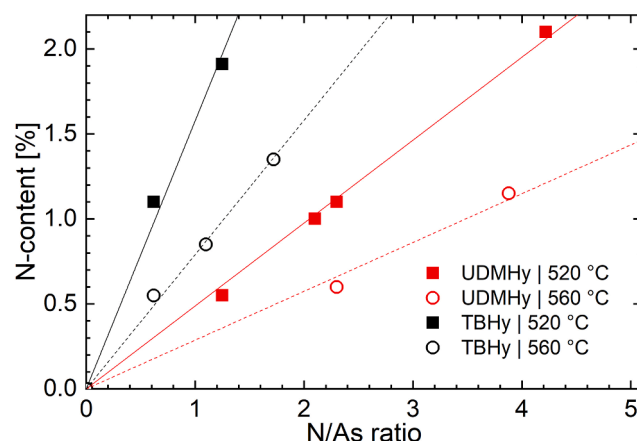


Fig. 2. N-content in GaNAs as a function of the N/As gas phase ratio for two different temperatures and both UDMHy and TBHy N precursors. All samples were grown with TEGa and TBAs as gallium and arsenic precursors, respectively, As/III ratio of 5 and growth rate of  $1.5 \mu\text{m/h}$ . The lines serve only as a guide for the eyes.

products on the surface, leading to a high desorption probability.

For the dilute nitrides the growth temperature is known to have a significant impact on the N-incorporation. Fig. 2 shows the N-content as function of the N/As ratio for different N-precursors for growth temperatures of 520 °C and 560 °C. With increasing temperature, the N-incorporation decreases significantly, which is again in good agreement with the literature [8,9,14]. Initially this is a surprising result as higher temperatures should lead to higher incorporation in a metastable system. However, this effect is very small for GaNAs [9]. Hence, other mechanisms dominate the N-incorporation. Most likely an increased desorption of the reactive N-species causes a decrease in the N-incorporation in the material. For the presented data, the N-incorporation decreases by half when the temperature increases from 520 °C to 560 °C. Lowering the growth temperature to 500 °C did not change the N-content (not shown here). This might indicate that the desorption of the reactive N-species is not dominating the N-incorporation efficiency for lower temperatures.

The third investigated issue is the N-precursor influence on the N-incorporation efficiency. In Fig. 2, results of samples grown with TBHy and UDMHy are represented by black and red data points, respectively. The N-incorporation is approximately three times higher for TBHy compared to that for UDMHy for both temperatures, in good agreement with the literature [9,12]. The increased N-incorporation with TBHy can be explained by the different decomposition mechanisms of the N-precursors and losses of the active N-species in chemical reactions. Chemical pre-reactions of the precursors can be neglected in our reactor system, as will be discussed later.

TBHy starts to decompose at a lower temperature compared to UDMHy (200 °C and 350 °C) [9,10,15]. Both molecules crack at the N-N bond releasing a  $\text{NH}_2$ -radical. Additionally, a dimethylamine  $(\text{CH}_3)_2\text{N}$ -radical and a tertiarybutylamine  $(\text{CH}_3)_3\text{CNH}$ -radical are obtained from UDMHy and TBHy, respectively. The temperatures for the full precursor decomposition vary immensely in the literature. This indicates that the chosen reactor design has a strong influence. Our reactor design most closely resembles the used reactor of [10], where a full decomposition of UDMHy is reached at around 800 °C. Thus, the lower N-incorporation efficiency with UDMHy can be correlated to a more inefficient precursor decomposition.

A further homogenous decomposition of the amine-radicals seems unlikely at the low temperatures and with the short interaction time allowed in this study. Further reactions of the  $\text{NH}_2$ -radicals with the amine-radical producing stable amines can lower the active N-species significantly [12]. These reactions are likely to happen in the gas-phase and on the surface. The size of the tertiarybutylamine may show a steric hindrance to the reaction, leading to higher N-concentration for TBHy. These free-radical reactions might also explain the significantly higher N-incorporation with Hydrazin reported by [12], because no ammine radicals are formed during the decomposition.

### 3.1.2. As/III ratio

Apart from the temperature, the V/III ratio is known to have a significant influence on the composition of the ternary materials. The absolute V/III ratio strongly depends on the high N-precursor partial pressure, which is used to control the N content in a linear fashion for low N contents (Fig. 2). However, the TBAs partial pressure remained so far unchanged. Hence, a series of experiments to study the As/III ratio influence in the N-incorporation was performed. The results obtained from the analysis are shown in Fig. 3. There, the N-content versus the N/As gas phase ratio for different N-precursors and As/III ratios is plotted. Changing the As/III ratio from 5 to 10 has essentially no effect in the N-incorporation. This is in contrast to prior studies conducted with TMGa, TBAs and DMHy in a horizontal atmospheric pressure reactor, for which increasing the As/III ratio led to a lower N-incorporation [13]. This difference can be explained by pre-reactions of the precursors, which are minimized in our reactor due to the lower reactor pressure (100 mbar) and the short interaction time. Additionally, the presented data shows

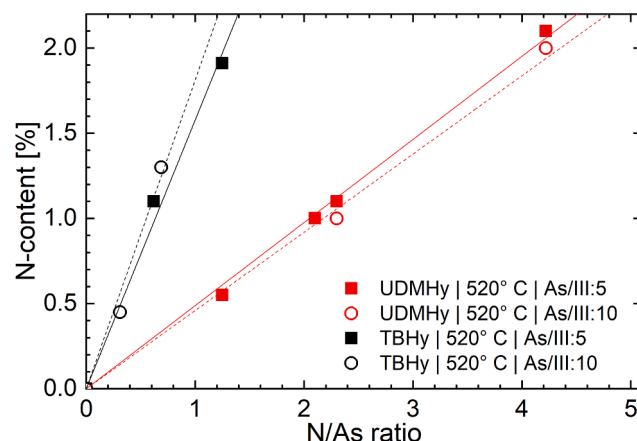


Fig. 3. N-content as a function of the N/As ratio in the gas phase for different N precursors, growth temperatures and As/III ratios. All samples were grown with TEGa and TBAs as gallium and arsenic precursors, respectively, at a temperature of 520 °C and a growth rate of 1.5  $\mu\text{m/h}$ . The lines are only a guide for the eyes.

that the N-incorporation depends only on the N/As-ratio, rather than on the absolute molar flow of the N-precursor.

### 3.1.3. Ceiling to susceptor distance and growth rate

To further investigate the influence of gas-phase and surface effects on the N-incorporation we studied the influence of the distance between gas inlet and wafer surface and the growth rate.

The specific design of the Carius showerhead reactor allows the study of another growth parameter which can be used to test the influence of the interaction time in the gas-phase on the N-incorporation. By changing the gap between the reactor ceiling and susceptor, the precursor concentration and temperature gradient can be adjusted. The effect can be seen in Fig. 4, where the N-content and the growth rate as a function of the gap-distance is plotted. Reducing the gap from 16 to 8 mm increases the growth rate by 40%, from 1.4  $\mu\text{m/h}$  to around 2.0  $\mu\text{m/h}$ . However, the N-concentration remains unaltered. The higher growth rate can be correlated to the increased precursors' concentration gradient. Hence, the reduced gap corresponds to a situation with a larger gap and an increased molar flow.

These results further indicate that the N-incorporation is dominated by surface rather than gas phase effects, like pre-reactions between the

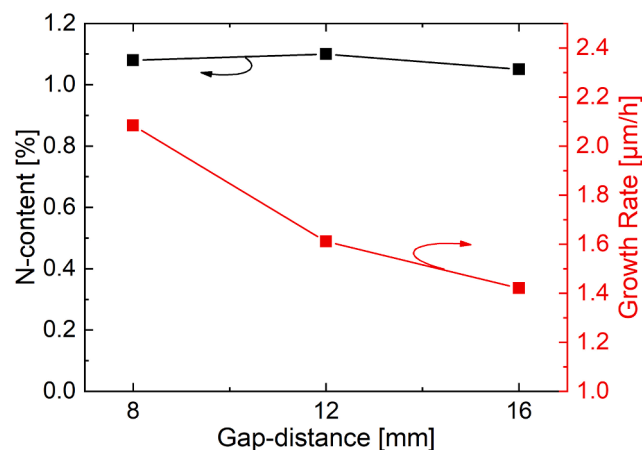


Fig. 4. N-content and growth rate for GaNAs as a function of the gap between wafer and showerhead. All samples were grown with TBHy, TEGa and TBAs as nitrogen, gallium and arsenic precursors, respectively, at a temperature of 520 °C with an As/III ratio of 5 and a N/As ratio of 0.62. The lines serve only as guides for the eyes.

precursors. Additionally, the results suggest that the N-incorporation is not desorption limited in the investigated regime, as the higher growth rates would otherwise increase the N-concentration in the crystal.

To verify this hypothesis we studied the influence of the growth rate on the N incorporation. In Fig. 5, the N-content versus the N/As gas phase ratio for different growth rates is plotted. The N-incorporation increases by a factor of four when the growth rate is increased from 0.2  $\mu\text{m/h}$  to 1.0  $\mu\text{m/h}$ . However, increasing the growth rate further, from 1.0  $\mu\text{m/h}$  to 2.0  $\mu\text{m/h}$ , shows no influence on the N-incorporation. In the literature different trends have been observed. Kurtz et al. observed a similar trend, showing an enhanced N-incorporation for increasing growth rates, which saturates at growth rates above 6  $\mu\text{m/h}$  on substrates with an offcut of  $2^\circ$  [16]. In contrast, Volz et al. reported no influence of the growth rate on the N-incorporation for GaNAs on substrates with offcut of less than  $0.5^\circ$  [9]. However, as it will be discussed later, given a few different growth conditions, their data could, in fact, indicate a similar behavior.

It should be mentioned that the surface of a thicker GaAs-layer grown at such low rates ( $\leq 0.2 \mu\text{m/h}$ ) shows a significant amount of step-bunching. However, the surface of the 10 nm thin GaNAs films shows no enhanced step-bunching for the low growth rates.

According to our data, the observed behavior of the decreased N-incorporation at low growth rates indicates an enhanced N-loss-mechanism. We assume that in this case, the effective N-radical concentration is limited by desorption from the terrace surface. For higher growth rates a sufficient amount of N-H<sub>2</sub>-radicals reaches the steps and the desorption at the surface steps dominates the N-incorporation. The desorption probability at the steps does not depend on the growth rate. Hence, a saturation of the N-incorporation efficiency with respect to the growth rate occurs. Such a mechanism would depend on the terrace width. This might explain the different results in literature.

The observed results of Fig. 5 fit well with those of Fig. 4, which had already indicated no influence of the growth rate on the N incorporation for the growth rate range between 1.4  $\mu\text{m/h}$  to 2  $\mu\text{m/h}$ . Furthermore, the results discussed in the last two sections clearly show that chemical pre-reactions can be neglected in this study. Finally, it should be emphasized that even if the higher growth rates show no influence on the N-incorporation, the majority of the active N-species desorbs from the surface.

### 3.1.4. Ga-precursor

After analysing the effects of the N-precursor, the influence of the Ga-precursor on the N-incorporation was studied. The most common Ga-precursors, TMGa and TEGa, were used to grow GaNAs under otherwise unaltered conditions. These experiments were performed at 520

and 560  $^\circ\text{C}$  at a growth rate of 1.5  $\mu\text{m/h}$  with an As/III-ratio of 5. The Ga molar flow was tuned to achieve similar growth rates. Three sets of samples were grown, two using TBHy and one with UDMHy to verify a possible influence of the N-precursor. The N-content versus the N/As-gas-phase-ratio for these experiments is plotted in Fig. 6. Using TEGa as Ga-precursor results in a higher N-incorporation for the lower growth temperature compared to TMGa. For 560  $^\circ\text{C}$  this difference vanishes.

At these growth conditions the growth efficiency of TMGa shows a strong temperature dependence, requiring an increase of the TMGa molar flow to maintain a growth rates of 1.5  $\mu\text{m/h}$  compared to the growth at 560  $^\circ\text{C}$ . This indicates a decomposition limited growth process with TMGa in our reactor system, while the growth efficiency with TEGa remains constant (not shown here).

In the literature reduced N-incorporation efficiency with TMGa compared to TEGa has also been reported [17]. The explanation given was an enhanced decomposition of AsH<sub>3</sub> in the presence of TEGa. This seems unlikely as more As would lower the effective N/As-ratio, leading to less N in the crystal.

Additionally, we exclude gas-phase reactions to explain the decreased N-incorporation with TMGa at lower temperatures, as we previously showed that pre-reactions of the precursors can be neglected in our reactor system. Thus, the formation of adducts between the Ga- and N-precursors [11] can be disregarded in our reactor.

We believe that the increased presence of the methyl-groups with TMGa at lower temperatures causes the decrease of the N-incorporation. Under these conditions, methyl-groups have a lower desorption rate and there are less H-radicals present from the decomposition of TBAs to saturate the methyl-groups. Additionally, more methyl-groups are still attached to the Ga-atoms at the lower temperatures, indicated by the decreased growth efficiency. Once a methyl-group, free or attached to a Ga-atom, and the active NH<sub>2</sub>-species interact, it is very likely that stable armines are formed [12]. The methylamines would desorb and lower the effective population of the active N on the surface. Consequently, the N-incorporation efficiency is decreased.

This latest result shows that even if the gas-phase reaction can be neglected in our system, clearly the chemical reaction on the surface might have a significant impact.

### 3.1.5. Reactor system comparison

Finally, we compare our results of the N-incorporation with the those of the literature [9,12]. While Volz et al. used a low pressure (50 mbar), horizontal MOVPE reactor system (AIX 200-GFR-reactor), Ptak et al. grew their samples in a custom made atmospheric pressure reactor.

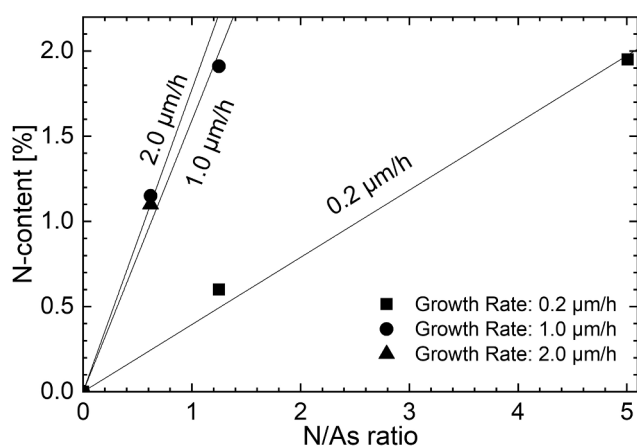


Fig. 5. N-content in GaNAs as function of the N/As ratio for different growth rates. All samples were grown with TBHy, TEGa and TBAs as nitrogen, gallium and arsenic precursors, respectively, at a temperature of 520  $^\circ\text{C}$  and an As/III ratio of 5. The lines serve only as a guide for the eyes.

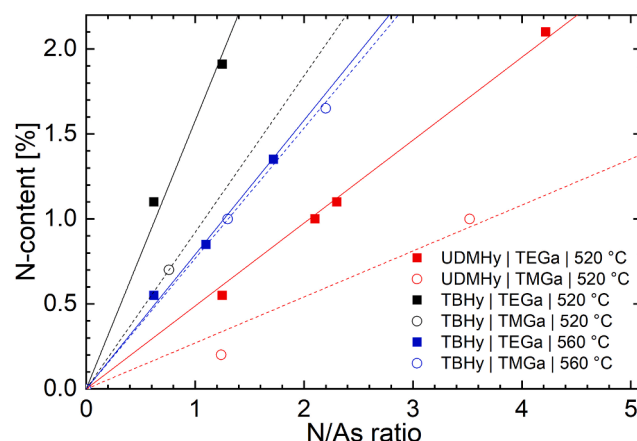


Fig. 6. N-content in GaNAs as a function of the N/As-gas-phase-ratio using UDMHy, as well as TBHy as nitrogen precursor, TEGa and TMGa as gallium precursors for 520  $^\circ\text{C}$  and 560  $^\circ\text{C}$  growth temperatures. All samples were grown with TBAs as arsenic precursor and an As/III-ratio of 5 at a growth rate of 1.5  $\mu\text{m/h}$ . The lines serve only as a guide for the eyes.



Using these reactor systems for comparison, we can cover a wide range of growth conditions.

For an appropriate comparison of the N-incorporation efficiency of the different reactor systems, similar growth conditions are required. Fig. 7 shows the N-incorporation in GaNAs grown with TEGa and TBAs for different temperatures, N-precursors and reactor systems. The first observation is the overall trend, present in all reactor systems, of enhanced N-incorporation with the use of TBHy versus UDMHy.

The results from Volz et al. can be compared with our lower temperature data. Interestingly, the showerhead shows a higher N-incorporation, although it has a shorter time for the precursor decomposition. This is surprising, as Volz and co-workers reported that the N-incorporation with UDMHy is limited by the precursor decomposition [9]. The higher reactor pressure in our system might compensate for the lower prereaction time. The comparison of the Ptak et al. data with our results obtained with the higher growth temperature reveals a similar or higher N-incorporation efficiency.

Here, it should be mentioned that temperature offsets in the systems might have an impact on the reported data absolute values. However, the comparison between the reactor systems shows the suitability of the showerhead for commercial applications due to its high N-incorporation efficiency.

At this point the effects of different parameters on the growth of GaNAs in the showerhead reactor can be summarized. The most important parameter is the N/As-ratio followed by the temperature and the N- and Ga-precursors choice. The absolute flow and the growth rate show a minor influence within a suitable growth window, as applied in this study. Gas-phase pre-reactions can be neglected while surface reactions are present. Apart from the chemical reactions (stable armine formation and insufficient decomposition) the N-incorporation in GaNAs is primarily dominated by the N/As ratio, temperature and strain (not shown here). We assume that the effective N/As-ratio at surface steps is a key-parameter, but the N incorporation is still influenced by temperature and strain. Less strain and lower temperatures lead to a lower desorption, increasing the N-content in the crystal. The maximum solubility of N in GaAs was not part of this study and should depend mainly on the growth temperature and strain state of the crystal.

### 3.2. C background doping

After the detailed study of the N-incorporation, the second part of this work covers the C-incorporation in GaNAs determined by SIMS. We tried to identify the source of C and investigate the influence of the

growth parameters on its incorporation. Thus, a path to lower the C-background in the dilute nitrides was identified.

For the first experiment, two layers with TBHy as N-precursor were grown, one with TEGa and the other with TMGa to study the Ga-precursor influence on the C-background doping. Furthermore, two more layers with TMGa and TEGa as precursors under similar conditions were grown, both using UDMHy as the N-precursor. All samples exhibit the same N-concentration of around 1%. These experiments are designed to understand the influence of Ga- and N-precursors on the background doping and to identify whether carbon is primarily supplied by either Ga-or N-precursors.

In Fig. 8 the C-background doping is plotted for each N-precursor using TMGa and TEGa as gallium precursors. The results show that there is a negligible influence of the N-precursors on the C-background doping, as expected [18]. However, the C-background doping strongly depends on the Ga-precursor used, presenting around  $10^{16}$  and  $10^{18}$  atoms/cm<sup>3</sup> when TEGa or TMGa are used, respectively, in agreement with previously reported data [19].

These results clearly identify TMGa as the significant source of C-background in GaNAs. The methyl-groups from the radical-decomposition of TMGa form stable armines with NH<sub>2</sub> and are incorporated into the crystal. On the other hand, TEGa is decomposed via a  $\beta$ -hydride elimination, forming the stable Ethene which is not incorporated. The invariance of the used N-precursor indicates that the UDMHy precursor is not completely decomposed at temperatures experienced during growth. Hence, its methyl-groups remain bonded to the second N-atom preventing their incorporation into the crystal.

After identifying the main C background source, we have investigated the influence of further growth parameters on its incorporation. Most of these experiments were performed with TEGa as the Ga-precursor to achieve the minimal possible C-background. First, we have verified the influence of the N-concentration in the crystal. Fig. 9 shows the carbon background in GaNAs grown with UDMHy, TBAs and TEGa at 520 °C for 1 and 2% N in the crystal. The carbon background rises by a factor of 4, from  $1.5 \times 10^{16}$  to  $6 \times 10^{16}$  atoms/cm<sup>3</sup>, when the N-concentration is doubled, a trend already reported in the literature [18,19]. As the very stable N-C-bond is responsible for the high C-background in dilute nitrides, such result was naturally expected. Hence, any comparison of the effects of different growth parameters on the C incorporation should be made between samples with a similar N-concentration. We decided to use GaN<sub>0.01</sub>As<sub>0.99</sub> as a standard composition for further experiments in this section, unless otherwise stated. Furthermore, devices requiring higher N-concentration will have the drawback of higher C-background levels.

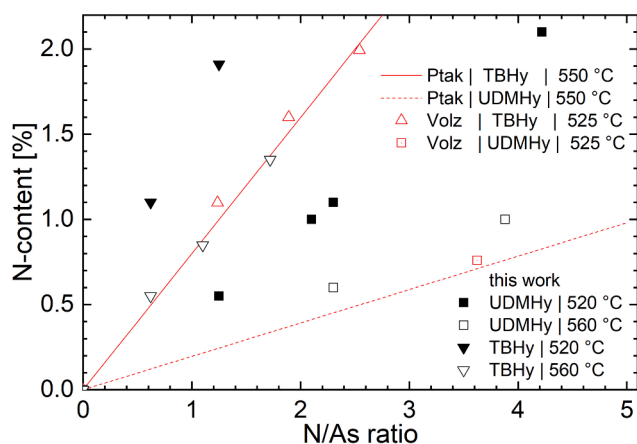


Fig. 7. N-content in GaNAs as a function of the N/As gas phase ratio for samples for which UDMHy, as well as TBHy have been used as nitrogen precursors [9,12]. Samples were grown with TEGa and TBAs as gallium and arsenic precursors. The red lines are a best fit for Ptak et al. data. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

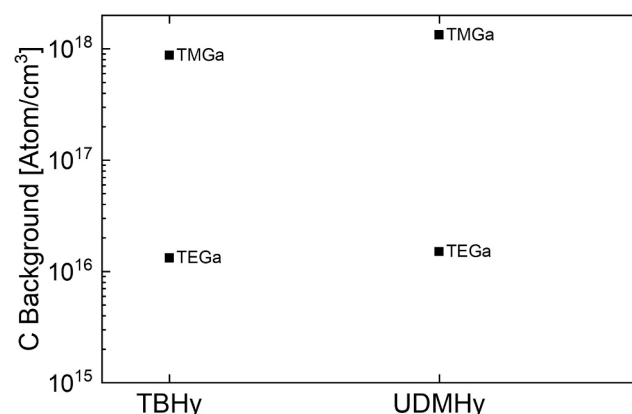
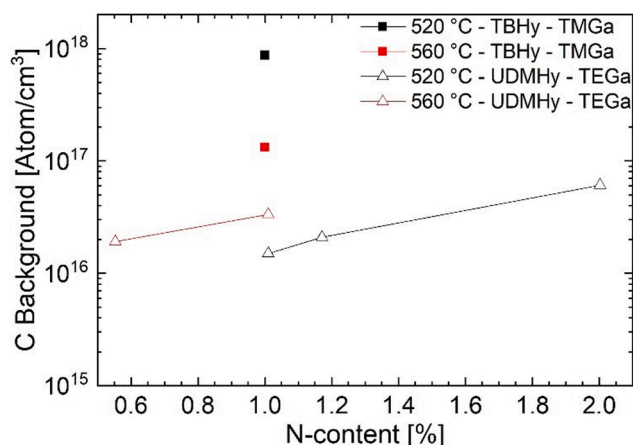


Fig. 8. C-background doping in GaNAs determined by SIMS for UDMHy, as well as TBHy, as nitrogen precursors and TEGa and TMGa as gallium precursors. All samples were grown with TBAs as arsenic precursor at a temperature of 520 °C with an As/III-ratio of 5. The N/As ratio was adjusted to achieve the similar N-concentration in the solid phase of 1%.



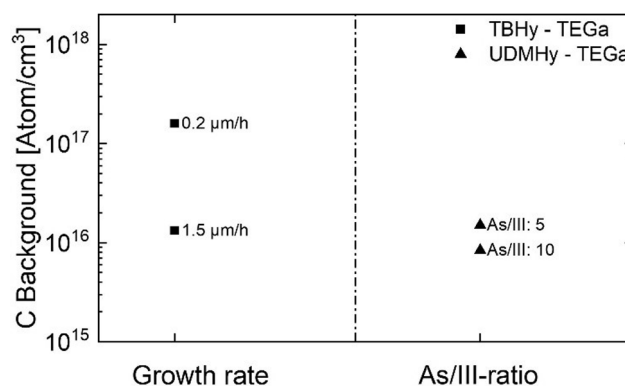
**Fig. 9.** Influence of the growth conditions on the C-background doping determined by SIMS. The used precursors are indicated in the plot. All samples were grown with an As/III-ratio of 5.

The second growth parameter tested was the growth temperature. Fig. 9 includes the C-background for GaNAs grown at 520 and 560 °C. For the first set of samples TEGa and UDMHy were used as precursors, while the second set was grown with TMGa and TBHy. While there is only a small offset in the C-concentration for the TEGa-samples, in the case of the TMGa ones, the C-background decreased by an order of magnitude when the growth temperature was raised. The former effect can be explained by an enhanced decomposition of UDMHy, releasing methyl-groups and thus increasing the C-background. Additionally, the TEGa precursors may start to decompose via a radical-decomposition, releasing ethyl-radicals, which could, in turn, be incorporated into the crystal. However, a small offset in the N-concentration may also explain part of the revealed differences in C-incorporation. This first result is in good agreement with the literature [18]. Volz et al. observed a constant C-background in GaNAs when using TEGa for growth temperatures above 500 °C.

The significant decrease in the C-background for higher growth temperatures when TMGa is used as a Ga precursor must be correlated to its strong temperature dependent decomposition in this reactor. As mentioned in the last section, increasing the growth temperature in this range increases the decomposition of TMGa, thus lowering the methyl-group density attached to the Ga-atoms on the surface. Interactions between the N-H<sub>2</sub>-radicals and these methyl-groups will enhance the C-incorporation and the N-desorption simultaneously. While the C-background is still one order of magnitude higher compared to samples for which TEGa was used, the reduction is significant. This trend fits well with the results from the reduced N-incorporation with TMGa compared to TEGa for a growth temperature of 520 °C, which is not present for higher growth temperatures (Fig. 6).

The following investigated parameter was the As/III-ratio. Fig. 10 shows the C-background for GaNAs grown at 520 °C with UDMHy and TEGa at As/III-ratios of 5 and 10. To achieve similar N-concentrations we kept the N/As-ratio constant. Doubling the As/III-ratio reduces the C-background to half. This could be correlated to the increased H-radical-density originating from TBAs, which might saturate methyl-radicals. In the literature a similar trend has been observed [18]. Based on the obtained results, we can neglect contaminations in the N-precursor as the cause for the C-background, as a doubled amount of UDMHy leads to a lower C-concentration. A similar observation has been made when TBHy was used to grow GaInNAs (not shown here). The absence of contaminations in the N-precursors allows a fair comparison of samples grown with different N/As-ratios.

The last investigated parameter is the growth rate. As we showed in Fig. 5 lowering the growth rate by an order of magnitude significantly reduces the N-incorporation. Hence, we compared the C-background in



**Fig. 10.** C-background concentration, determined by SIMS, as a function of growth rate (using TBHy as the nitrogen precursor) and As/III ratio (using UDMHy as the nitrogen precursor). TEGa was used as the gallium precursor. If not stated otherwise, all samples are grown at 520 °C with an As/III-ratio of 5 and with a N-concentration of 1% in the solid.

GaNAs grown at 0.2 and 1.5 μm/h with TBHy and TEGa at 520 °C (Fig. 10). The SIMS analysis revealed an order of magnitude higher C-concentration for the slowly grown sample. To our knowledge, there has been no investigation in the literature on the influence of this growth parameter on the C-incorporation. Contamination from the N-precursor due to the increased N/As-ratio, necessary to reach a similar N-concentration (see Section 3.1.3), can be excluded as a cause for this high C-background, as mentioned in the last paragraph. However, we cannot exclude C-background contaminations from the reactor due to increased growth time. As mentioned before the GaNAs-surface of thick layers may show step-bunching. The increased roughness may lead to an enhanced C-incorporation, as it has been reported that higher C-background is present in samples with more surface steps [20]. However, this result shows clearly that low growth rates should be avoided to ensure a high N- and a low C-incorporation.

In summary, methyl-groups from the TMGa-precursor could be identified as the major C-source while the N-precursors have a minor influence. The C-background is proportional to the N-concentration in the crystal, as expected, and the optimal growth temperature for low C-densities depends on the used precursors. Furthermore, to achieve lower C-backgrounds, high As/III-ratios and growth rates are beneficial. While the use of TEGa significantly reduces the C-background, the lowest measured C-density was still in the upper range of 10<sup>15</sup> atoms/cm<sup>3</sup>. Increasing the N-concentration in the GaNAs-film will further increase the C-background.

#### 4. Conclusion

While dilute nitrides offer great possibilities for applications due to their unique bandgap and lattice parameter properties, the understanding of the growth with MOVPE of the metastable material remains a challenge. In our analysis it was found that the low N-incorporation efficiency of GaNAs is dominated by the desorption of the active N-species. The N-incorporation is a linear function of the N/As-ratio, allowing a precise control of low N-concentrations. The growth rate, gas-phase-ratio and gap between the showerhead and the GaAs wafer surface show no or minor influence, indicating the absence of gas-phase pre-reactions in the chosen reactor system. The decreased N-incorporation with higher growth temperatures is due to the enhanced desorption of the N-species. Differences in the N incorporation depending on the N-precursor can be explained by the pyrolysis and radical reaction between NH<sub>2</sub> and carbon groups of the precursors. The latter is also responsible for a lower N-incorporation when TMGa is used as compared to TEGa, highlighting the presence of surface reactions.

Understanding the growth of GaNAs allowed an in-depth

investigation of the unintentional C-incorporation in the material. Methyl-groups from TMGa have been identified as the major source of Carbon, while the N-precursors show only a minor influence. The strong N-C-bond causes an increase of C-background with higher N-content. Therefore, only samples with similar N-contents can be directly compared. Lower growth temperatures when using TEGa led to lower carbon doping densities, while this trend changes for TMGa, showing the influence of the pyrolysis of the precursor. If decent growth rates ( $\geq 1 \mu\text{m/h}$ ) and high As/III-ratio are maintained, low C-background levels below  $9 \times 10^{15} \text{ atoms/cm}^3$  could be achieved. In this case more H-radicals from the TBAs pyrolysis passivate the Carbon-radicals and hinder their incorporation.

### CRediT authorship contribution statement

**J.E. Ruiz:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization. **D. Lackner:** Validation, Resources, Writing - review & editing, Project administration, Funding acquisition. **P.L. Souza:** Writing - review & editing, Supervision. **F. Dimroth:** Resources, Writing - review & editing, Supervision, Funding acquisition. **J. Ohlmann:** Validation, Formal analysis, Investigation, Writing - original draft, Visualization, Project administration, Funding acquisition.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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